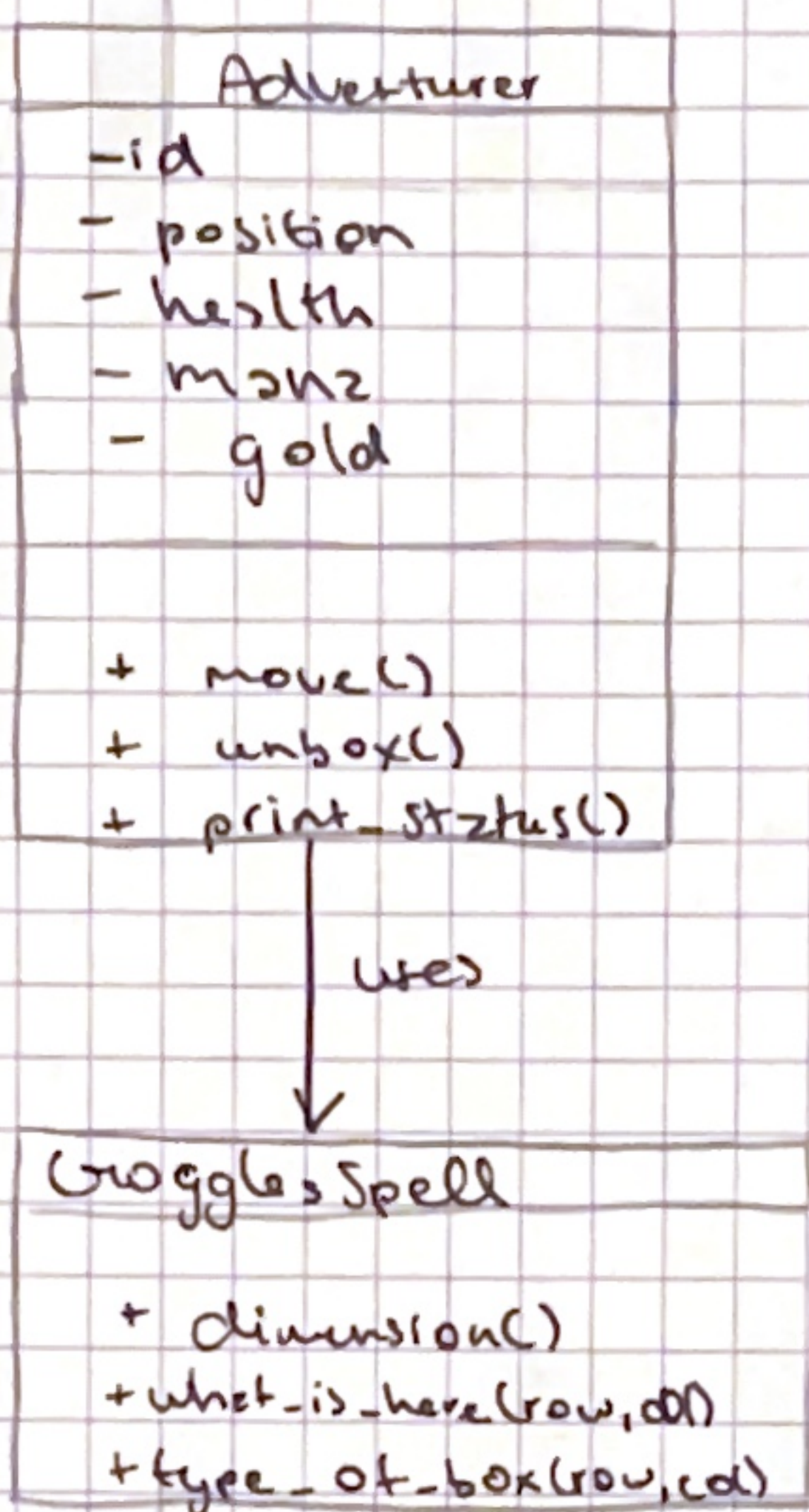
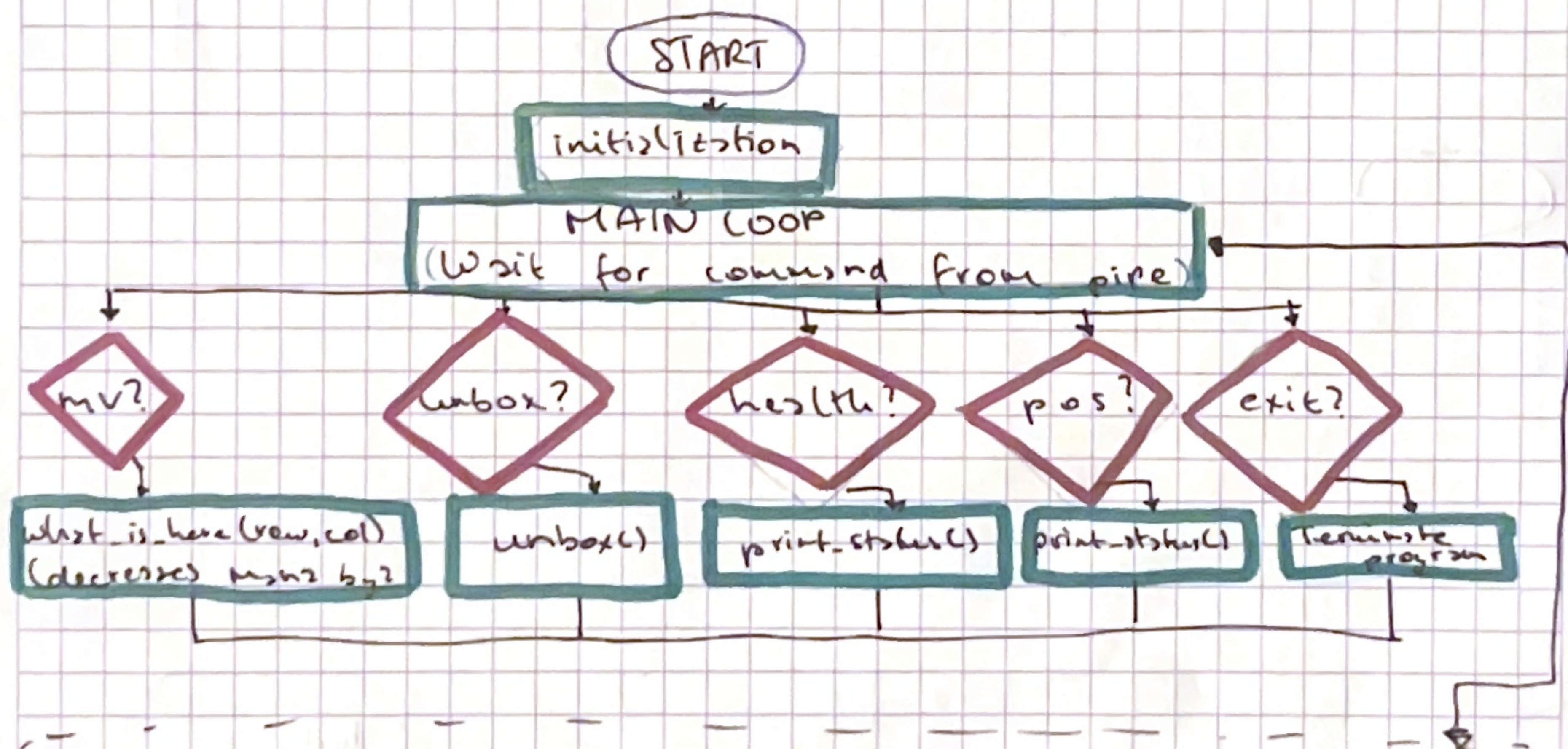


Exercise 1: adventurer.py

Class diagram:



Flowchart



Assign known Signal Handlers

SIGUSR1

SIGUSR2

SIGQUIT

SIGTSTP

SIGCONT

SIGINT

SIGALRM

health == 1
every 2 seconds

Mentor

Parse arguments

Load party and validate

Create N adventurers and pipes

Connect signals

While

input()

if print

YES

Print map

elif NO

mv <dirs> <cd>

YES

if

There's collision

YES

Print "collision"

elif NO

adv pipe: "mv"

FIFO: MOVE

if HIT

YES

os.kill(SIGINT)

health, pos, unbox

YES

Write command to adventurer

elif NO

rebuild/destruct

YES

Send signal to PID

else

exit

YES

Print positions, gold and final state

Send exit command to childs (pipes)

'N' os.wait() (for each child)

End of Program

Methods:

parse-arguments
load-party-and-validate

Mentor

(uses GoogleSpell)

adventurer1

adventurer2

adventurerN

Variables:

mentor.name
dungeon.file
party.file
dungeon.dims
dungeon.map
adventurers
channels
opened-boxes
total-gold
fifo.out.dm
fifo.in.dm

SIGUSR1/2

send signal to all the childs PID's

print information and actualize total gold

resume execution

SIGINT

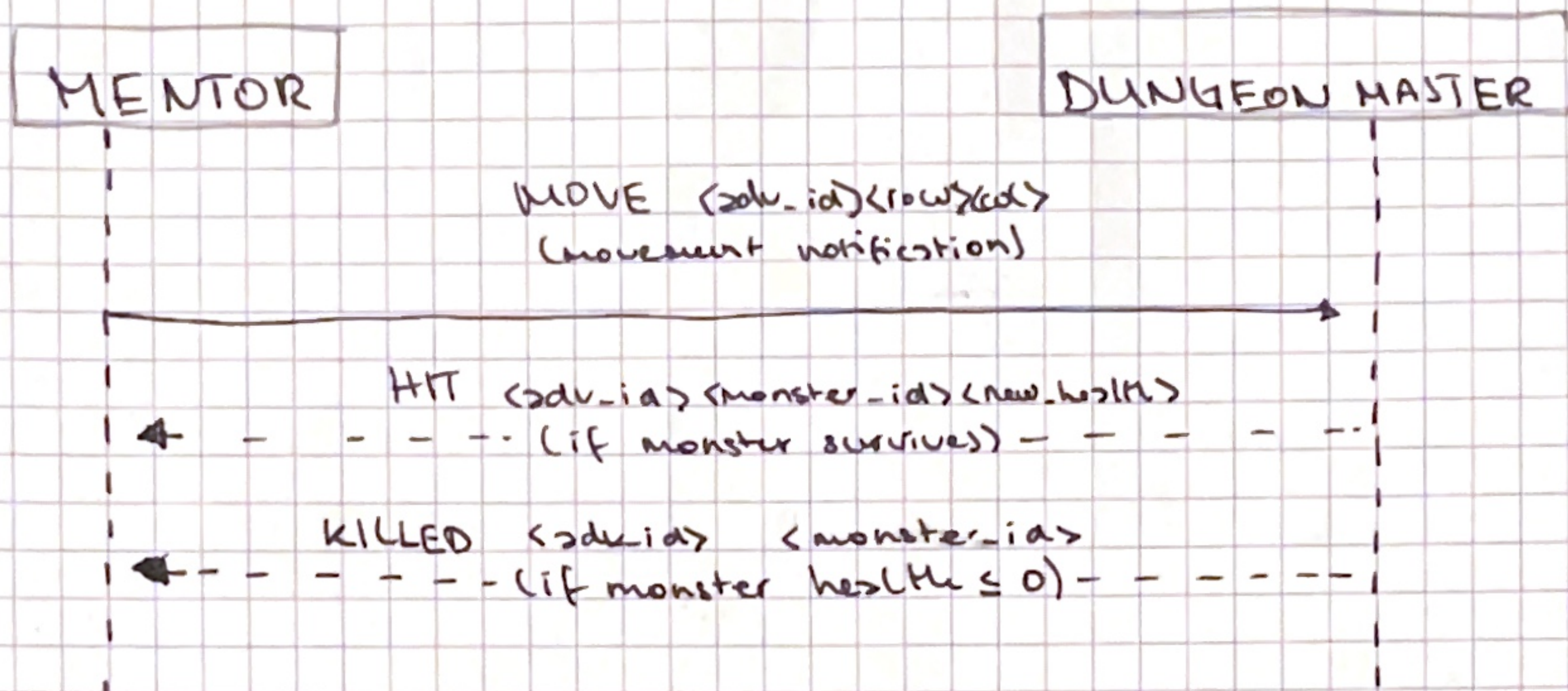
print final position

kill all adventurers and do wait()

end of program

Exercise 3: dm.py

Protocol Sequence Diagram



Message/ Command	FORMAT	DESCRIPTION	Response
MOVE	Move <adv_id><row><col>	Adventurer want to cell	HIT or killed
HIT	HIT <adv_id><monster_id> <new_health>	Monster hit, still alive	None
KILLED	KILLED <adv_id><monster_id>	Monster defeated	None

Flowchart:

